

Exploring the Design Space of Adaptive Explanations in Automated Vehicles

Philipp Asteriou, Ambika Shahu, Andreas Riegler, Philipp Wintersberger

IT:U Interdisciplinary Transformation University Austria

philipp.asteriou@it-u.at

When should an automated vehicle explain itself or stay silent?

Explanation needs change with context and experience [1–3]. We built a simulator to explore how vehicles adapt their explanations across situations, users, and time.

01 A platform for adaptive explanations

An incident is a scripted traffic event that prompts a vehicle response, such as braking, changing lanes, or stopping.



Motion and VR passenger experience

Unity driving environment with repeatable urban and highway incidents.



Situated feedback inside the vehicle

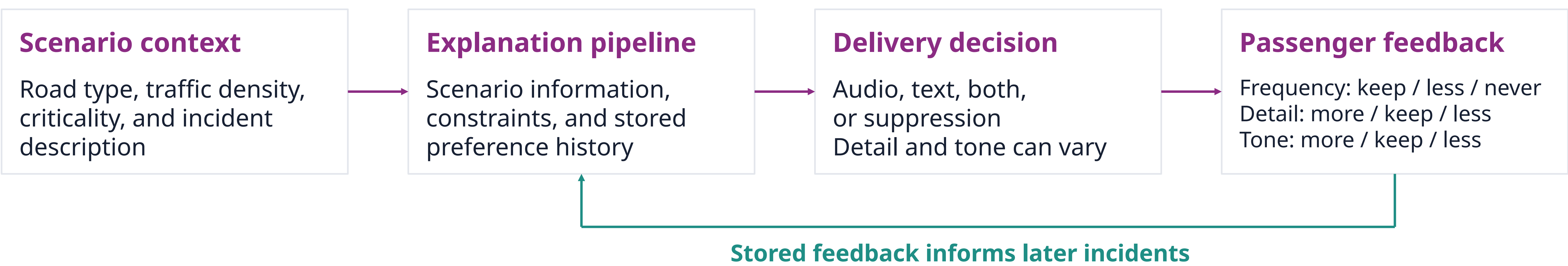
Passengers adjust frequency, detail, and tone after each incident.



Example: a blocked road

Event: a truck blocks the road. Response: the vehicle stops because dense traffic prevents passing.

02 How the adaptation loop works



Prototype constraints: use supplied scenario facts, avoid jargon, limit length, and log outputs.

03 Open design questions

- Silence** Should the vehicle indicate that it has suppressed an explanation?
- Safety** When should safety needs override a preference for fewer explanations?
- User control** How can users request explanations or revise their preferences?

Current scope and next steps

Research prototype. Empirical evaluation is planned.

LLMs support prototyping. Incorrect or misleading explanations remain a concern [4]. Safety rules and validation require further development. Next: compare proactive, on-demand, and hybrid strategies. Extend scenarios and user controls.

Selected references

[1] Koo et al. (2015). Why did my car just do that? Explaining semi-autonomous driving actions to improve driver understanding, trust, and performance. IJIDeM 9, 269–275. doi:10.1007/s12008-014-0227-2

[2] Kim et al. (2023). What and When to Explain? On-road Evaluation of Explanations in Highly Automated Vehicles. PACM IMWUT 7(3), Article 104. doi:10.1145/3610886

[3] Wintersberger et al. (2020). Explainable Automation: Personalized and Adaptive UIs to Foster Trust and Understanding of Driving Automation Systems. AutomotiveUI '20, 252–261. doi:10.1145/3409120.3410659

[4] Sadeghi et al. (2024). Explaining the Unexplainable: The Impact of Misleading Explanations on Trust in Unreliable Predictions for Hardly Assessable Tasks. UMAP '24, 36–46. doi:10.1145/3627043.3659573